

LAB-2024-001

DIGITAL FORENSICS INVESTIGATION REPORT

DFIR Investigation — Windows XP Forensic Dataset

Case Number	LAB-2024-001
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Classification	Educational / Unclassified
Tools Used	Volatility 3, Autopsy, Wireshark, PE Studio, YARA
Evidence Sources	Memory Dump: seized.vmem Disk Image: seized.E01 Network Capture: capture.pcap

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1. Executive Summary

A Digital Forensics and Incident Response (DFIR) investigation was conducted on a Windows XP forensic dataset consisting of a memory image, forensic disk image, and network packet capture. The primary objective of the investigation was to identify evidence of malware activity, persistence mechanisms, suspicious network communications, and indicators of compromise.

Memory analysis was performed using Volatility 3, disk analysis using Autopsy, network analysis using Wireshark, and static executable analysis using PE Studio. Additional activities included YARA rule creation, IOC extraction, and MITRE ATT&CK technique mapping.

The investigation identified normal user activity, installed software artifacts, browser activity, network communications, and executable files. No definitive evidence of active malware infection, process injection, persistence mechanisms, or command-and-control activity was identified within the scope of the investigation.

This investigation successfully demonstrated a complete DFIR workflow including evidence acquisition, forensic analysis, IOC documentation, threat mapping, and professional reporting.

2. Evidence Acquisition

The following evidence items were collected and analyzed during the investigation:

Evidence Item 1 — Memory Image

- File: seized.vmem
- Purpose: Memory forensics analysis using Volatility 3

Evidence Item 2 — Forensic Disk Image

- File: seized.E01
- Purpose: Disk artifact analysis using Autopsy

Evidence Item 3 — Network Packet Capture

- File: capture.pcap
- Purpose: Network traffic analysis using Wireshark

Evidence integrity was maintained throughout the investigation. File hashes were generated and recorded to ensure evidence authenticity. All analysis activities were conducted within an isolated virtual laboratory environment to prevent contamination.

3. Investigation Methodology

The investigation followed a standard DFIR methodology encompassing the phases listed below:

Phase 1 — Evidence Acquisition

- Memory image collection
- Disk image collection
- Network capture acquisition

Phase 2 — Evidence Preservation

- Hash verification of all evidence items
- Evidence organization and chain-of-custody documentation

Phase 3 — Memory Analysis

- Process enumeration and scanning
- Command-line argument review
- Registry hive identification
- Memory injection detection
- DLL path analysis

Phase 4 — Disk Analysis

- File system review and artifact enumeration
- Browser artifact analysis
- Deleted file recovery
- USB device artifact review

Phase 5 — Network Analysis

- DNS query investigation
- HTTP traffic analysis
- IOC extraction from packet capture

Phase 6 — Static Analysis

- PE structure review
- Hash generation and metadata analysis
- Certificate examination

Phase 7 — Detection Engineering

- YARA rule creation and validation
- MITRE ATT&CK framework mapping

Phase 8 — Reporting

- Documentation of findings and analysis
- Conclusions and recommendations

4. Memory Forensics Analysis

Objective: Identify active processes, network connections, command-line activity, memory injection, and registry artifacts from the seized memory image.

Tool: Volatility 3

Evidence File: seized.vmem

4.1 System Information

Operating System: Windows XP SP3

System Root: C:\WINDOWS

Architecture: 32-bit

Processors: 2

Memory Capture Time: 2009-12-04 00:37:49 UTC

Note: The capture timestamp reflects the system clock recorded in memory and may not represent the actual acquisition time.

4.2 Process Analysis

The following processes were identified as noteworthy during analysis:

- cmd.exe (PID 3124) — Interactive command shell launched from explorer.exe. Required command-line argument review.
- firefox.exe (PID 2684) — Web browser active at time of acquisition.
- thunderbird.exe (PID 764) — Email client active at time of acquisition.
- mdd_1.3.exe (PID 3336) — Memory acquisition tool used to capture the RAM image.

4.3 Process Scan Analysis

A comparison between pslist and psscan outputs was performed to identify hidden or unlinked processes.

- No hidden processes were identified.
- All significant processes observed in psscan were also present in pslist.
- No evidence of DKOM-style (Direct Kernel Object Manipulation) process hiding was observed.

4.4 Command-Line Analysis

cmd.exe (PID 3124)

- Standard Windows command prompt.
- No suspicious command-line arguments observed.

firefox.exe (PID 2684)

- Executed from Program Files\Mozilla Firefox.
- Command-line arguments consistent with normal browser launch.

thunderbird.exe (PID 764)

- Executed with the "-mail" parameter.
- Consistent with normal email client usage.

mdd_1.3.exe (PID 3336)

- Memory acquisition utility.
- Command: Z:\mdd_1.3.exe -o z:\charlie-2009-12-03.ram
- Confirms memory image acquisition activity.

4.5 Malfind Analysis

The malfind plugin was used to identify suspicious memory regions with PAGE_EXECUTE_READWRITE (RWX) permissions and potential embedded PE headers.

- No MZ executable headers were identified in memory regions flagged by malfind.
- Multiple PAGE_EXECUTE_READWRITE regions were detected in: csrss.exe (PID 924), winlogon.exe (PID 948), lsass.exe (PID 1004), AVGIDSAgent.exe (PID 1888), avgwdsvc.exe (PID 1964), avgtray.exe (PID 3936), soffice.bin (PID 1148), firefox.exe (PID 2684).
- Given the Windows XP SP3 environment and the presence of antivirus software, these findings are inconclusive and may represent legitimate executable memory allocations rather than malicious code injection.
- No direct evidence of injected PE files was identified.

4.6 DLL Analysis

DLL load paths were reviewed to detect suspicious loading locations.

- No DLLs loaded from Temp directories.
- No DLLs loaded from AppData or other non-standard executable locations.
- Thunderbird loaded components from: C:\Documents and Settings\Charlie\Application Data\Thunderbird\Profiles\...
- This loading pattern is consistent with normal Thunderbird profile and extension storage behavior.
- No suspicious DLL loading activity was identified.

4.7 Registry Analysis

Registry hives were successfully identified within the memory image.

System Hives Identified

- SYSTEM
- SOFTWARE
- SAM
- SECURITY
- DEFAULT

User Hives Identified

- Charlie — NTUSER.DAT
- LocalService — NTUSER.DAT
- NetworkService — NTUSER.DAT

4.8 Memory Forensics Conclusion

Memory forensic analysis of seized.vmem using Volatility 3 identified normal user processes including a web browser, email client, command shell, and memory acquisition utility. Process comparison between pslist and psscan revealed no evidence of process hiding or DKOM-based manipulation.

Malfind analysis identified PAGE_EXECUTE_READWRITE memory regions in several processes; however, no embedded PE files or injected executable code was confirmed. DLL analysis revealed no suspicious loading patterns. Registry hives were successfully identified.

Overall, memory analysis did not reveal definitive indicators of malware activity or system compromise within the scope of this investigation.

5. Disk Forensics Analysis

Objective: Examine disk artifacts including installed applications, run program history, web activity, deleted files, USB devices, and browser history.

Tool: Autopsy

Evidence File: seized.E01

5.1 Installed Programs

The following applications were identified as installed on the system:

- Mozilla Firefox — Web browser
- Mozilla Thunderbird — Email client
- OpenOffice — Productivity suite
- AVG Antivirus — Security software
- Adobe Reader — Document viewer
- Python 2.6 — Scripting runtime

5.2 Web Downloads

The following download artifact was recovered from browser history:

- python-2.6.4 installer — Downloaded from www.python.org
- Source URL: <http://www.python.org/ftp/python/2.6.4/python-2.6...>
- Activity is consistent with legitimate software installation.

5.3 Interesting Items

No items of immediate investigative concern were identified during this review phase. Artifact analysis was consistent with normal workstation usage.

5.4 Extension Mismatch Analysis

A review of file extension mismatches was conducted to identify files potentially renamed to disguise their true type. No significant extension mismatches indicative of malicious activity were identified during the reviewed artifacts.

5.5 Run Programs

No unauthorized or suspicious startup programs were identified in the examined artifacts. Installed software was consistent with expected user applications.

5.6 USB Device Analysis

A review of USB device history artifacts was performed. The following observations were made:

- No evidence of USB flash drive attachment was identified.
- Observed USB device entries appear consistent with standard workstation peripherals (keyboard and mouse).
- No unauthorized external storage device connections were identified.

5.7 Deleted File Analysis

A total of 1,139 deleted file entries were recovered during analysis. The following representative artifacts were observed:

OpenOffice Configuration

- org.openoffice.FirstStartWizard.dat

AVG Antivirus Components

- avgdm90_pro_0700_027.pack

Temporary Files

- ~DF83C.tmp, ~DFBA73.tmp, ~DFBACA.tmp, ~DFBB50.tmp
- ~DFBB5E.tmp, ~DFC1EA.tmp, ~DFC257.tmp, ~DF91E5.tmp

Installer-Related Files

- sp1033.MST

No obviously malicious executable files (.exe), batch files (.bat), scripts (.vbs), or malware payloads were immediately identified in the reviewed deleted file artifacts. Further review is recommended pending full Autopsy ingest completion.

5.8 Web History Analysis

The following websites were identified within browser history artifacts:

- openoffice.org — Software download activity
- mozilla.com — Browser-related activity
- microsoft.com / update.microsoft.com — System and application updates
- msn.com — General web activity
- dell.com / support.dell.com — Hardware support activity
- sun.com — Java or development-related activity
- serving-sys.com — Advertisement or web tracking activity

Web browsing activity appears predominantly related to software updates, browser downloads, and system maintenance. No obvious malware-hosting domains were identified during the reviewed entries.

5.9 Disk Forensics Conclusion

Disk forensic analysis conducted on the E01 forensic image using Autopsy revealed evidence of normal user activity. Web history primarily contained vendor and update-related websites. USB artifacts showed only standard peripherals, with no evidence of external storage device connections.

Deleted file recovery identified numerous temporary and application-related files, but no clearly malicious executables, scripts, or unauthorized software artifacts were found. No suspicious persistence mechanisms or unauthorized startup programs were observed.

Based on available evidence, the disk image primarily reflects normal workstation usage. No definitive indicators of compromise were identified during the scope of this analysis.

6. Network Forensics Analysis

Objective: Analyze captured network traffic to identify suspicious DNS activity, HTTP communications, and potential command-and-control or data exfiltration indicators.

Tool: Wireshark

Evidence File: capture.pcap

Host IP Identified: 10.8.15.133

DNS Server: 10.8.15.4

6.1 DNS Analysis

Multiple DNS requests were observed between the host system (10.8.15.133) and DNS server (10.8.15.4). Observed query targets included:

- lafontainebleu.org
- microsoft.com
- officeclient.microsoft.com
- msftncsi.com

No domain generation algorithm (DGA)-style domains, fast-flux DNS behavior, or obviously malicious query targets were identified during DNS analysis.

6.2 HTTP Analysis

HTTP traffic analysis revealed the following activity from the host system:

Microsoft Connectivity Check

- GET /connecttest.txt — Destination: 23.205.110.48
- Purpose: Standard Microsoft Network Connectivity Status Indicator (NCSI) check.

Repeated HTTP GET Requests

- Destination: 199.232.210.172
- Resource: /filestreamingservice/files/
- Multiple HTTP 206 Partial Content responses observed.

6.3 HTTP POST Analysis

Three HTTP POST requests were identified originating from the host system (10.8.15.133) directed to remote server 72.5.43.29.

Total POST Requests: 3

Source: 10.8.15.133

Destination: 72.5.43.29

Observed Responses: HTTP/1.1 200 OK; HTTP/1.1 400 Bad Request

The POST requests indicate outbound communication from the host system. Based on available packet data, the purpose of this communication could not be conclusively determined. No direct evidence of data exfiltration was identified during packet review.

6.4 Network Forensics Conclusion

Network traffic analysis using Wireshark did not reveal definitive evidence of malicious command-and-control communications or active data exfiltration within the analyzed packet capture.

DNS activity was limited to legitimate Microsoft services and the domain lafontainebleu.org. HTTP traffic included standard connectivity checks and file streaming activity. Three outbound HTTP POST requests to 72.5.43.29 were observed, but insufficient packet data was available to conclusively determine their purpose.

Conversation analysis revealed limited communication volume and no clear beaconing patterns. No large file transfers, exploit traffic, or confirmed malware downloads were identified. Overall, observed traffic appears largely consistent with normal application and system communications.

7. Static Analysis

Objective: Perform static analysis of a recovered executable to evaluate PE structure, metadata, hash values, entropy, and embedded indicators.

Tool: PE Studio

Analyzed File: Wireshark.exe

7.1 File Information

File Name: Wireshark.exe

File Type: 64-bit Portable Executable (PE)

File Size: 18,667,688 bytes

Version: 4.6.8

File Description: Wireshark

Original File Name: Wireshark.exe

7.2 Hash Analysis

SHA-256:

99DC13045CC75BA9FEAEB8E9C56A1BE0E2C22DEAC15B474A435469B33E6D76B1

MD5: 5DD9D2DB30F70063BFCAB99450417CE4

7.3 PE Studio Findings

- Valid PE executable structure detected.
- Executable identified as a GUI-based application.
- Entropy value: 7.327 — possibly indicative of compression or packing, though common in legitimately packed software.
- PE Studio did not identify obvious malicious indicators during basic review.
- No suspicious imports, embedded URLs, mutex names, or registry persistence indicators were observed.

7.4 Certificate Analysis

- PE Studio reported an expired certificate associated with the executable.
- An expired certificate alone does not constitute evidence of malicious activity.

7.5 Static Analysis Conclusion

Static analysis of the examined executable was performed using PE Studio. The file appears to be a legitimate network protocol analysis application. Metadata, PE structure, entropy values, and embedded information were reviewed.

No evidence of malicious functionality, process injection APIs, suspicious network indicators, or persistence mechanisms was identified during the scope of this analysis. The file appears consistent with a legitimate software installation and was used for educational static analysis purposes.

8. YARA Rule Creation

Objective: Develop a custom YARA detection rule based on findings from static analysis to demonstrate signature-based detection engineering.

Rule Name: LAB001_Malware_Detection

8.1 Detection Logic

The YARA rule was developed to identify executable files containing characteristics observed during static analysis. The rule performs the following checks:

- Verifies the presence of a valid PE executable (MZ header).
- Searches for strings identified during static analysis.
- Uses conditional logic to determine whether a file satisfies the defined detection criteria.

Observed Strings Used

- "Wireshark"
- "WiresharkDevelopmentTeam"

The full YARA rule is provided in Appendix B.

8.2 YARA Validation

The rule was successfully created and reviewed for syntax correctness. The exercise demonstrated the process of creating custom detection signatures based on forensic findings and static analysis artifacts.

8.3 YARA Conclusion

YARA provides an effective method for detecting files based on known characteristics, metadata, strings, and behavioral indicators. The rule developed during this investigation serves as a proof-of-concept demonstrating signature creation and validation within a DFIR workflow.

9. Indicators of Compromise (IOCs)

The following indicators of compromise were extracted and documented during the investigation. These IOCs are provided for reference and threat intelligence purposes.

Indicator Type	Value	Context
File Hash (SHA-256)	99DC13045CC75BA9FEAEB8E9C56A1BE0E2C22DEAC15B474A435469B33E6D76B1	Wireshark.exe — analyzed executable (PE Studio)
File Hash (MD5)	5DD9D2DB30F70063BFCAB99450417CE4	Wireshark.exe — analyzed executable (PE Studio)
IP Address	10.8.15.133	Host system observed in network capture
IP Address	72.5.43.29	Destination of outbound HTTP POST requests
IP Address	23.205.110.48	Destination of HTTP connectivity test traffic
Domain	lafontainebleu.org	DNS queries observed during network analysis
Domain	www.python.org	Download artifact identified in Autopsy
URL	http://www.python.org/ftp/python/2.6.4/python-2.6...	Web download record in Autopsy
Process	firefox.exe (PID 2684)	Active browser process — memory analysis

Process	thunderbird.exe (PID 764)	Active email client — memory analysis
Process	cmd.exe (PID 3124)	Interactive command shell — memory analysis
Process	mdd_1.3.exe (PID 3336)	Memory acquisition utility

Negative Findings

- No suspicious registry persistence artifacts identified.
- No mutexes identified during analysis.
- No suspicious scheduled tasks identified.

10. MITRE ATT&CK Mapping

The following MITRE ATT&CK techniques were mapped to observed evidence. Only techniques directly supported by available forensic evidence are included.

10.1 Techniques Observed

Technique ID	Technique Name	Evidence
T1071.001	Application Layer Protocol: Web Protocols	HTTP GET and POST communications observed between 10.8.15.133 and external IPs during Wireshark analysis.
T1071.004	Application Layer Protocol: DNS	DNS queries to 10.8.15.4 observed for lafontainebleu.org and Microsoft-related services.
T1204.002	User Execution: Malicious File	Web download artifacts and software installer records identified in Autopsy indicating user-initiated software execution.
T1027	Obfuscated Files or Information	PE Studio reported entropy value of 7.327 on analyzed executable, possibly indicating compression or packing. No malicious obfuscation confirmed.

10.2 Techniques Reviewed But Not Observed

The following techniques were considered during analysis but are not supported by the available evidence:

Technique ID	Technique Name	Rationale
T1055	Process Injection	No injected PE headers or anomalous memory regions confirmed.
T1055.012	Process Hollowing	No evidence of hollow process structures identified.
T1547.001	Registry Run Keys / Startup Folder	No suspicious registry persistence artifacts identified.
T1053.005	Scheduled Task/Job: Scheduled Task	No suspicious scheduled tasks identified.
T1003	OS Credential Dumping	No credential dumping tools or artifacts identified.

11. Recommendations

The following recommendations are based solely on findings identified during the investigation scope:

- Complete remaining Autopsy artifact review — Conduct a full review of remaining disk artifacts after Autopsy ingest reaches 100% completion. Additional artifacts may provide further investigative context.
- Monitor identified network indicators — The IP addresses 72.5.43.29 and 199.232.210.172, and the domain lafontainebleu.org, should be reviewed against current threat intelligence databases and monitored for future activity.
- Deploy YARA-based detection signatures — Implement the developed YARA rule within endpoint detection tooling to improve identification of executable files sharing characteristics with those analyzed.
- Enable centralized logging and monitoring — Deploy endpoint detection and response (EDR) tooling and centralized log management to improve visibility into system activity, network connections, and process execution.
- Maintain forensic readiness procedures — Establish and maintain documented evidence preservation procedures, including regular log retention policies and incident response playbooks.
- Conduct periodic IOC and threat intelligence reviews — Regularly review extracted IOCs against updated threat intelligence feeds to identify emerging threats associated with observed indicators.
- Maintain software update and patching discipline — Ensure all systems, particularly legacy operating systems, receive security patches promptly. Windows XP SP3 is an end-of-life platform and presents inherent risk.

12. Conclusion

Scope

A comprehensive digital forensic investigation was conducted on a Windows XP forensic dataset comprising a memory image, disk image, and network packet capture. The investigation encompassed memory forensics, disk forensics, network traffic analysis, static executable analysis, YARA rule development, IOC extraction, and MITRE ATT&CK mapping.

Evidence Reviewed

- Memory Image: seized.vmem
- Disk Image: seized.E01
- Network Capture: capture.pcap
- Executable: Wireshark.exe (static analysis subject)

Tools Used

- Volatility 3 — Memory forensics
- Autopsy — Disk forensics
- Wireshark — Network traffic analysis
- PE Studio — Static executable analysis
- YARA — Detection rule development

Major Findings

- Memory analysis identified active processes (cmd.exe, firefox.exe, thunderbird.exe, mdd_1.3.exe) consistent with normal user activity. No process hiding or memory injection was confirmed.
- Disk analysis revealed normal user and application activity. No malicious executables, unauthorized persistence mechanisms, or suspicious startup programs were identified.
- Network analysis identified DNS and HTTP communications. Three HTTP POST requests to 72.5.43.29 were observed but could not be conclusively attributed to malicious activity.
- Static analysis of Wireshark.exe confirmed a legitimate executable. An expired certificate and elevated entropy were noted but do not independently indicate malicious intent.

Final Assessment

Based on the evidence reviewed within the scope of this investigation, no definitive indicators of active malware infection or system compromise were identified. Analyzed artifacts were largely consistent with normal workstation activity. The investigation successfully demonstrated the complete DFIR workflow and provided practical experience in forensic acquisition, artifact analysis, IOC extraction, threat mapping, YARA development, and professional forensic reporting.

13. Investigation Limitations

- **Educational Lab Environment** — The investigation was conducted within a controlled educational laboratory environment using publicly available forensic datasets. Findings may not reflect real-world incident conditions.
- **Time Constraints** — Due to the investigation timeline, certain advanced analysis activities were not completed within scope, including full Autopsy artifact review pending 100% ingest completion.
- **Hardware Limitations** — Resource constraints within the virtual laboratory environment may have affected tool performance and the completeness of certain analysis phases.
- **Scope Limitations** — External threat intelligence enrichment, correlation with threat feeds, and in-depth malware reverse engineering were outside the defined scope of this investigation.
- **Legacy Platform Constraints** — The Windows XP SP3 memory image is not supported by Volatility 3's netscan plugin, limiting the ability to extract network connection artifacts directly from memory.
- **Future Investigation Areas** — Complete review of remaining deleted file artifacts, correlation of identified IP addresses with threat intelligence databases, and full Autopsy ingest completion are recommended as follow-on activities.